

Le Chang, PhD

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Research Centre, Montreal Heart Institute, 5000 rue Bélanger, Montreal, Quebec H1T 1C8, Canada

researchgate.net/profile/Le-Chang-4 · Permanent Resident of Canada

Research Focus

My research uses human genetic variation as a natural experiment to separate cause from consequence among the organ-level and molecular factors that shape aging and age-related disease, and to turn those causal signals into predictive biomarkers, drug targets and pharmacogenomic insight for precision health. I combine statistical genetics — multi-ancestry and sex-stratified genome-wide association, genetic correlation, colocalization and Mendelian randomization — with multi-omics integration of the proteome, metabolome and transcriptome, including molecular QTLs that link variants to drug-target and drug-metabolizing genes. My current domain is the relationship between systemic organ function (kidney, liver, metabolism) and neurodegeneration, where I identify shared causal variants and prioritize actionable proteins and metabolites for risk stratification and therapeutic repurposing. I build the open-source platforms and reproducible pipelines that make these genomic analyses usable by clinicians and researchers.

Academic Positions

- **Postdoctoral Research Fellow**, Research Centre, Montreal Heart Institute, and Faculty of Medicine, Université de Montréal, Montreal, Canada *September 2024 – present*
- **Postdoctoral Research Fellow**, Center for Precision Health, McWilliams School of Biomedical Informatics, UTHealth Houston, Houston, USA *September 2023 – August 2024*

Education and Training

- **PhD**, Human Genetics (Bioinformatics), Department of Human Genetics, McGill University, Montreal, Canada *May 2023*
Supervisor: Jianguo (Jeff) Xia · NSERC CREATE MATRIX Scholarship
Dissertation: Network-based approaches towards understanding genetics of metabolism and regulatory interactions
- **MBI**, Biomedical Informatics, School of Computing and the Department of Biomedical and Molecular Sciences, Queen's University, Kingston, Canada *May 2019*
Supervisor: Qingling Duan, Computational Genomics Laboratory
Project: Genome-wide association study of human milk oligosaccharides in the CHILD Cohort Study
- **BSc**, Nutrition and Food Science, Faculty of Agricultural, Life & Environmental Sciences, University of Alberta, Edmonton, Canada *June 2015*
Graduated with Distinction; Dean's List and First-Class Standing
- **BEng**, Bioengineering, Northwest A&F University, Yangling, China *June 2015*
"2+2" dual-degree program with the University of Alberta

Professional Experience

- **Leadership Trainee**, Maple Leaf Foods Inc., Canada *June 2015 – April 2018*
Hired into the company's leadership trainee program; progressed through Production Supervisor (2015–2016) and Business Process Analyst (2016–2017) to Supply Planning Analyst (2017–2018). Demand forecasting and inventory analytics; managed the recipe database for 11 manufacturing facilities; supervised a 20-person production team.

Funding

- **CIHR Fellowship**, Canadian Institutes of Health Research: Unraveling the genetic architecture and causal pathways of hematuria: a comprehensive multi-ancestry meta-analysis (my role: **principal applicant**; \$140,000) *2025–2027*
- **Postdoctoral Fellowship Award**, Heart and Stroke Foundation of Canada (my role: **principal applicant**; \$30,000; awarded and declined in favour of the CIHR Fellowship) *2025–2026*
- **CPRIT BIG-TCR Postdoctoral Fellowship**, Cancer Prevention and Research Institute of Texas, Biomedical Informatics, Genomics and Translational Cancer Research Training Program: "Decoding the role of viral transcriptional regulators in cancer" (my role: **principal applicant**; USD \$52,704) *2024*
- **Approved data access**, Canadian Longitudinal Study on Aging, application 2501015: "Multi-omics Mendelian randomization approaches to identify causal modifiable risk factors for dementia" (three-year access; principal applicant: Sarah A. Gagliano Taliun; my role: **lead analyst**) *2025 – present*

Scholarships and Awards

- **CIHR Fellowship**, Canadian Institutes of Health Research *2025–2027*
- **Postdoctoral Fellowship Award**, Heart and Stroke Foundation of Canada (declined) *2025–2026*
- **CPRIT BIG-TCR Postdoctoral Fellowship** *2024*
- **MANA Early-Career Travel Award**, Metabolomics Association of North America *September 2022*
- **ISMB 2020 Fellowship Award**, International Society for Computational Biology *June 2020*
- **NSERC CREATE MATRIX Scholarship**, NSERC CREATE MATRIX Training Program *2019–2023*
- **Graduate Excellence Award**, Department of Human Genetics, McGill University *May 2019*
- **AllerGen Conference Travel Award**, AllerGen NCE Inc. *January 2019*
- **Graduation with Distinction**, University of Alberta *June 2015*
- **Dean's List and First-Class Standing**, University of Alberta *2014, 2015*
- **Ronald A. Chisholm Undergraduate Scholarship**, Canadian Meat Science Association *May 2015*
- **Ian Morrison Undergraduate Leadership Award**, University of Alberta *January 2015*
- **Macgregor Smith Public Speaking Competition**, Third Place, University of Alberta *January 2015*
- **Mary (Davidson) Howard Bar None Award**, University of Alberta *October 2014*
- **Student Challenge Competition**, First Place Team, Canadian Institute of Food Science and Technology *August 2014*
- **Hilda Wilson Memorial Volunteer Recognition Award**, University of Alberta *February 2014*
- **NWAFU–ALES Undergraduate Transfer Program Award**, University of Alberta *May 2013*

Publications (n 14)

Three ESI Highly Cited Papers (Web of Science); miRNet 2.0 has been cited more than 900 times.

Summary of publications since 2019	
	Quantity
Peer-reviewed journal articles	11
(Co-)First author	6
Co-author	5
Book chapters (first author)	2
Published conference abstracts (first author)	1

* Equal contribution

Peer-reviewed journal articles

1. **L. Chang***, S. Gawhary*, L. Maameri, W. Belbellaj, F. Lona-Durazo, S.A. Gagliano Taliun (2026) Genome-wide cross-trait analysis to compare genetic architecture between Parkinson's disease and kidney-related traits. *Nature Communications*. Published 18 September 2026. Doi: 10.1038/s41467-026-77920-3
2. J. Bellavance*, H. Xiao*, **L. Chang**, M. Kazemi, S. Wickramasinghe, A.J. Mayhew, P. Raina, P. VandeHaar, D. Taliun, S.A. Gagliano Taliun (2026) Exploring and visualizing stratified genome-wide association study results with PheWeb 2. *Nature Genetics*. 58(2):233–235. Doi: 10.1038/s41588-025-02469-8. PMID: 41555088
3. M. Zuo*, **L. Chang***, N. Neale, L. Maameri, S. Gawhary, F. Lona-Durazo, S.A. Gagliano Taliun (2025) The relationship between kidney health and neurodegenerative diseases. *Brain*. 148(8):2616–2630. Doi: 10.1093/brain/awaf113. PMID: 40120077; Podcast: open.spotify.com/episode/0OdP5vSjzd7nMwLEetqsR
4. C. Citu, **L. Chang**, A.M. Manuel, N. Enduru, Z. Zhao (2025) Identification and catalog of viral transcriptional regulators in human diseases. *iScience*. 28(3):112081. Doi: 10.1016/j.isci.2025.112081
5. A. Ambalavanan, **L. Chang**, J. Choi, Y. Zhang, S.A. Stickley, Z.Y. Fang, K. Miliku, B. Robertson, C. Yonemitsu, S.E. Turvey, P.J. Mandhane, E. Simons, T.J. Moraes, S.S. Anand, G. Paré, J.E. Williams, B.M. Murdoch, G.E. Otoo, S. Mbugua, E.W. Kamau-Mbuthia, E.W. Kamundia, D.K. Gindola, J.M. Rodriguez, R.G. Pareja, D.W. Sellen, S.E. Moore, A.M. Prentice, J.A. Foster, L.J. Kvist, H.L. Neibergs, M.A. McGuire, M.K. McGuire, C.L. Meehan, M.R. Sears, P. Subbarao, M.B. Azad, L. Bode, Q. Duan (2024) Human milk oligosaccharides are associated with maternal genetics and respiratory health of human milk-fed children. *Nature Communications*. 15:7735. Doi: 10.1038/s41467-024-51743-6
6. **L. Chang***, J. Ewald*, F. Hui, S. Bayen, J. Xia (2024) A data-centric perspective on exposomics data analysis. *Exposome*. 4(1):osae005. Doi: 10.1093/exposome/osae005
7. **L. Chang**, G. Zhou, J. Xia (2023) mGWAS-Explorer 2.0: causal analysis and interpretation of metabolite–phenotype associations. *Metabolites*. 13(7):826. Doi: 10.3390/metabo13070826
8. **L. Chang**, G. Zhou, H. Ou, J. Xia (2022) mGWAS-Explorer: linking SNPs, genes, metabolites, and diseases for functional insights. *Metabolites*. 12(6):526. Doi: 10.3390/metabo12060526
9. Z. Pang, G. Zhou, J. Ewald, **L. Chang**, O. Hacariz, N. Basu, J. Xia (2022) Using MetaboAnalyst 5.0 for LC–HRMS spectra processing, multi-omics integration and covariate adjustment of global metabolomics data. *Nature Protocols*. 17(8):1735–1761. ESI Highly Cited Paper
10. Z. Pang, J. Chong, G. Zhou, D.A. de Lima Morais, **L. Chang**, M. Barrette, C. Gauthier, P.-É. Jacques, S. Li, J. Xia (2021) MetaboAnalyst 5.0: narrowing the gap between raw spectra and functional insights. *Nucleic Acids Research*. 49(W1):W388–W396. ESI Highly Cited Paper
11. **L. Chang**, G. Zhou, O. Soufan, J. Xia (2020) miRNet 2.0: network-based visual analytics for miRNA functional analysis and systems biology. *Nucleic Acids Research*. 48(W1):W244–W251. ESI Highly Cited Paper; more than 900 citations

Book chapters

12. **L. Chang**, J. Xia, Z. Zhao (2025) Computational approaches for studying signal transduction networks at cellular context. In: S. Ranganathan, M. Cannataro, A.M. Khan (eds) *Encyclopedia of Bioinformatics and Computational Biology*, 2nd edition. Elsevier. pp. 436–444.
13. **L. Chang**, J. Xia (2022) MicroRNA regulatory network analysis using miRNet 2.0. In: Q. Song, Z. Tao (eds) *Transcription Factor Regulatory Networks. Methods in Molecular Biology*. Springer. pp. 185–204.

Published conference abstracts

14. **L. Chang**, A. Ambalavanan, A. Lamri, B. Robertson, C. Yonemitsu, S. Turvey, P. Mandhane, A. Becker, T. Moraes, S. Anand, G. Paré, D. Lefebvre, M. Sears, P. Subbarao, L. Bode, M. Azad, Q. Duan (2019) Genome-wide association study of human milk oligosaccharides in the CHILD Study. *Allergy, Asthma & Clinical Immunology* (AllerGen 2019 Research Conference abstracts).

Thesis

L. Chang (2023) Network-based approaches towards understanding genetics of metabolism and regulatory interactions. PhD dissertation, *Department of Human Genetics, McGill University*.

Software and Research Resources

- **miRNet 2.0** (mirnet.ca) — lead developer and first author. Network-based visual analytics platform for microRNA functional analysis and systems biology, integrating miRNAs, genes, transcription factors, SNPs, lncRNAs and circRNAs across multiple species. Responsible for system architecture, database design, Java/R back end, interactive network visualization, deployment, tutorials and multi-year user support. More than 70,000 users; ESI Highly Cited Paper.
- **mGWAS-Explorer 1.0 and 2.0** (mgwas.ca) — lead developer and first author. Platforms linking SNPs, genes, metabolites, diseases and molecular QTL evidence for the interpretation of metabolite–phenotype associations; version 2.0 adds two-sample Mendelian randomization across more than 4,000 metabolites and literature-derived semantic triples, with an accompanying R package.
- **MetaboAnalyst 5.0** (metaboanalyst.ca) — contributing developer. Designed and implemented the data-driven network analysis module and rebuilt the underlying KEGG interaction database for a platform serving more than 500,000 users.
- **PheWeb 2** (<https://github.com/GaglianoTaliun-Lab/PheWeb2>) — contributor to an open-source platform for exploring and visualizing stratified genome-wide association study results.
- **Analysis pipelines** — reproducible, version-controlled workflows (Nextflow, SLURM) for multi-ancestry GWAS meta-analysis (REGENIE), LD score regression, local genetic correlation (LAVA), conjunctive FDR, colocalization and multivariable Mendelian randomization on Digital Research Alliance of Canada clusters.

Oral Presentations

Oral Presentations as Invited Speaker

1. International Young Scholars Forum, School of Public Health, Southeast University. "Deciphering genetic architecture of human metabolism and gene regulatory networks: implications for public health" (Nanjing, China; 26 December 2024)
2. International Young Scholars Forum, School of Public Health, Nanjing Medical University. "Deciphering genetic architecture of human metabolism and gene regulatory networks: implications for public health" (Nanjing, China; 21 December 2024)
3. International Young Scholars Forum, Institute of Future Agriculture, Northwest A&F University. "Multi-omics data integration in precision medicine: applications and prospects" (Virtual meeting; 28 November 2024)
4. Center for Precision Health Seminar in Precision Medicine, University of Texas Health Science Center at Houston. "Deciphering genetic architecture of human metabolism and regulatory interactions" (Houston, Texas; 2023)
5. Institute of Intelligent Healthcare, School of Artificial Intelligence, Nanjing University of Information Science and Technology. "Decoding genetic variants of molecular phenotypes and regulatory networks" (Virtual meeting; 7 September 2022)

6. McGill University – Jilin University Bioinformatics Workshop. "MicroRNA regulatory network analysis using miRNet 2.0" (Montreal, Quebec; 2022)
7. Nutraceuticals and Functional Foods Global Classroom Session, Universiti Malaysia Pahang. "Human milk oligosaccharides" (Virtual meeting; 2021)

Oral Presentations

8. Three Minute Thesis Competition, Queen's University. "Investigation of genetic modulators of human milk oligosaccharide concentrations in nursing mothers" (Kingston, Ontario; 2019)
9. AllerGen 2019 Research Conference, trainee oral presentation. "Genome-wide association study of human milk oligosaccharides in the CHILD Study" (Toronto, Ontario; January 2019)

Poster Presentations

Presenting author marked (presenter)

1. **L. Chang*** (presenter), S. Gawhary*, L. Maameri, W. Belbellaj, F. Lona-Durazo, S.A. Gagliano Taliun. "Genome-wide cross-trait analysis to compare genetic architecture between Parkinson's disease and kidney-related traits" American Society of Human Genetics (ASHG) Annual Meeting (Montreal, Quebec; October 2026; accepted)
2. J. Boulais-Richard, H. Xiao, **L. Chang**, P. Raina, D. Taliun, S.A. Gagliano Taliun. "PheWeb2.1: Exploring stratified GWAS and interaction results with the ability to easily append new summary statistics" American Society of Human Genetics (ASHG) Annual Meeting (Montreal, Quebec; October 2026; accepted)
3. **L. Chang*** (presenter), S. Gawhary*, L. Maameri, W. Belbellaj, F. Lona-Durazo, S.A. Gagliano Taliun. "Assessment of global and local genetic correlations provides insights on the kidney–brain axis" American Society of Human Genetics (ASHG) Annual Meeting (Boston, Massachusetts; October 2025)
4. S. Gawhary, **L. Chang**, L. Maameri, W. Belbellaj, F. Lona-Durazo, S.A. Gagliano Taliun. "Assessment of global and local genetic correlations between kidney-related traits and late-onset neurodegenerative diseases" Journée de la Recherche de l'Institut de Cardiologie de Montréal (Montréal, Québec; 5 June 2025)
5. M. Zuo, **L. Chang**, N. Neale, L. Maameri, S. Gawhary, F. Lona-Durazo, S.A. Gagliano Taliun. "A review of the relationship between kidney health and neurodegenerative diseases" Journée de la Recherche de l'Institut de Cardiologie de Montréal (Montréal, Québec; 5 June 2025)
6. M. Zuo, **L. Chang**, N. Neale, L. Maameri, S. Gawhary, F. Lona-Durazo, S.A. Gagliano Taliun. "Does kidney health influence dementia and neurodegeneration" VAST Training Program Conference (Toronto, Ontario; 27 May 2025)
7. S. Wickramasinghe, H. Xiao, J. Bellavance, **L. Chang**, S.A. Gagliano Taliun, D. Taliun. "Principal component analysis on the European subset from the Canadian Longitudinal Study of Aging" McGill Undergraduate Science Showcase (Montreal, Quebec; 2025)
8. **L. Chang** (presenter), G. Zhou, J. Xia. "mGWAS-Explorer 2.0: causal analysis and interpretation of metabolite–phenotype associations" 4th Annual Canadian Metabolomics Conference (Niagara-on-the-Lake, Ontario; June 2023)
9. **L. Chang** (presenter), G. Zhou, H. Ou, J. Xia. "mGWAS-Explorer: linking SNPs, genes, metabolites and diseases for functional insights" 4th Annual Conference of the Metabolomics Association of North America (MANA) (Edmonton, Alberta; September 2022)
10. **L. Chang** (presenter), G. Zhou, H. Ou, J. Xia. " mGWAS-Explorer: linking SNPs, genes, metabolites and diseases for functional insights " 2022 Ottawa Institute of Systems Biology Symposium (Cornwall, Ontario; May 2022)

11. **L. Chang** (presenter), G. Zhou, H. Ou, J. Xia. "mGWAS-Explorer: a visual analytics platform connecting metabolomics with genomics" 17th Annual Conference of the Metabolomics Society (Virtual meeting; June 2021)
12. Z. Pang, J. Chong, G. Zhou, D.A. de Lima Morais, **L. Chang**, M. Barrette, C. Gauthier, P.-É. Jacques, S. Li, J. Xia. "MetaboAnalyst 5.0: narrowing the gap between raw spectra and functional insights" 17th Annual Conference of the Metabolomics Society (Virtual meeting; June 2021)
13. A. Ambalavanan, **L. Chang**, J. Choi, A. Lamri, B. Robertson, C. Yonemitsu, S.E. Turvey, P.J. Mandhane, A.B. Becker, T.J. Moraes, S.S. Anand, G. Paré, D.L. Lefebvre, M.R. Sears, P. Subbarao, L. Bode, M.B. Azad, Q. Duan. "Genome-wide association study of human milk oligosaccharides in lactating mothers and interaction effects on lung health among infants in the CHILD Study" 69th Annual Meeting of the American Society of Human Genetics (ASHG), poster 2948 (Houston, Texas; October 2019)
14. **L. Chang** (presenter), G. Zhou, O. Soufan, J. Xia. "miRNet 2.0: network-based visual analytics for miRNA functional analysis and systems biology" 28th Conference on Intelligent Systems for Molecular Biology (ISMB 2020), Network Biology (NetBio) COSI poster session (Virtual meeting; July 2020) <https://www.youtube.com/watch?v=xyIDNN9tuiU>
15. A. Ambalavanan*, **L. Chang***, J. Choi, A. Lamri, B. Robertson, C. Yonemitsu, S. Turvey, P. Mandhane, A. Becker, T. Moraes, S. Anand, G. Paré, D. Lefebvre, M. Sears, P. Subbarao, L. Bode, M. Azad, Q. Duan. "GWAS of maternal human milk oligosaccharides demonstrates protective effects on preschool wheeze in the CHILD Study" 8th Annual Canadian Human and Statistical Genetics Meeting (Montebello, Quebec; June 2019)
16. **L. Chang** (presenter), H. Ou, O. Soufan, G. Zhou, J. Xia. "mGWAS-DB: elucidating genetically influenced metabolotypes for mGWAS" 8th Annual Canadian Human and Statistical Genetics Meeting (Montebello, Quebec; June 2019)
17. **L. Chang** (presenter), A. Ambalavanan, A. Lamri, B. Robertson, C. Yonemitsu, S. Turvey, P. Mandhane, A. Becker, T. Moraes, S. Anand, G. Paré, D. Lefebvre, M. Sears, P. Subbarao, L. Bode, M. Azad, Q. Duan. "Genome-wide association study of human milk oligosaccharides in the CHILD Study" AllerGen 2019 Research Conference (Toronto, Ontario; January 2019)
18. **L. Chang** (presenter), M. Ndagijimana, Z. Khiari, M. Betti. "Poultry glycosaminoglycans: a new source of dietary 'animal fibers' and candidate prebiotics?" Alberta Poultry Research Centre Annual General Meeting (Edmonton, Alberta; May 2014)

Teaching

At McGill University — Graduate level

- **Teaching Assistant**, BINF 531: Statistical Bioinformatics *Winter 2023*
Ran the hands-on tutorials and marked the analyses for this graduate course.
- **Teaching Assistant**, BTEC 501: Bioinformatics *Fall 2021*
Ran the hands-on tutorials and marked the analyses for this graduate course.

Workshops, tutorials and user training

- **Instructor**, McGill University – Jilin University Bioinformatics Workshop, Montreal *2022*
Taught a hands-on miRNet 2.0 session on microRNA regulatory network analysis to trainees from China.
- **Author of training materials**, miRNet and mGWAS-Explorer *2020 – 2023*
Online tutorials, documentation, slide decks and video tutorials for the two platforms (audience: more than 70,000 miRNet users), and a step-by-step protocol chapter in Methods in Molecular Biology (2022).

- **User support** 2020 – 2023
Answered user questions from students, clinicians and researchers by email and on the laboratory's online user forum; user questions fed back into the tutorials and into new platform features.

Supervision/Mentorship

Trainee co-supervision, Montreal Heart Institute / Université de Montréal (with S.A. Gagliano Taliun)

Master's-level researchers — cross-trait genetics of liver markers and neurodegenerative disease
2025 – present

Trainee co-supervision, University of Texas Health Science Center at Houston

Undergraduate summer intern, CPRIT BIG-TCR Summer Intern Program — deep-learning model for prediction of HTLV-1 viral integration sites
June – August 2024

Co-mentoring of undergraduate students, McGill University

Two undergraduate research students — community forum and RASA conversational agent for web-based bioinformatics platforms
June 2021 – June 2022

Peer mentoring, University of Alberta

Two international undergraduate students — academic guidance and referral to campus support services
May 2013 – September 2015

Certifications and Other Training

- Potential and Strengths Coach Course, Freedom Life Coach Limited, ICF CCE September 2025
- CIHR Sex- and Gender-Based Analysis Plus (SGBA+) online training modules, Canadian Institutes of Health Research September 2024
- Informatics and Statistics for Metabolomics, Canadian Bioinformatics Workshops June 2020
- Expanding Horizons Professional Development Certificate, Queen's University March 2019
- Ontario Summer School on High Performance Computing (Compute Ontario) August 2018
- Six Sigma White Belt, Maple Leaf Foods Inc. October 2015
- Competent Communicator, Toastmasters International March 2015
- ULead Leadership Program, University of Alberta December 2014

Professional Service

- **Youth Editor**, iMeta February 2025 – February 2028
- **Reviewer** for Brain Communications, Journal of Alzheimer's Disease, Scientific Reports and other journals 2024 – present
- **Member**, ASHG Volunteer Pool September 2024 – present
- **Adjudicator**, Canadian Undergraduate Conference on Healthcare, Queen's University November 2018
- **Organizer and host**, ALES Networking Career Workshop, Faculty of Agricultural, Life & Environmental Sciences, University of Alberta February 2014
- **VP Communications**, Nutrition and Food Science Students' Association, University of Alberta 2013 – 2015
- **Professional memberships**: American Society of Human Genetics (2018 – present); Metabolomics Association of North America (2022 – present); Global Organisation for Bioinformatics Learning, Education & Training, GOBLET (2018 – present); European Society of Human Genetics (2018); International Society for Computational Biology (2020 – present); International Common Disease Alliance (2019 – present); Golden Key International Honour Society (2014 – present); AllerGen NCE, student member (2018 – 2019)

Knowledge Translation and Outreach

- **Invited guest**, The Brain Podcast (official podcast of the journal Brain) — "Nephrology Meets Neurology: How Kidney Health Links to Neurodegeneration"; discussed the kidney–brain findings of the Brain 2025 review for clinicians, researchers and the public *June 2025*
- **Presenter**, Fun Speakers Toastmasters Club — public talks on food safety, genetically modified foods and alternative protein sources *October 2014 – May 2015*

Technical Skills

- **Statistical genetics**: GWAS (REGENIE, PLINK), multi-ancestry meta-analysis, global genetic correlation (LDSC), local genetic correlation (LAVA), conjunctural FDR, fine-mapping (SuSiE), colocalization, univariable and multivariable Mendelian randomization (TwoSampleMR, MVMR-cML-SuSiE), polygenic scores, sex-stratified and chromosome X analysis, molecular QTL (eQTL, mQTL, pQTL) integration
- **Multi-omics**: genomics, metabolomics, transcriptomics, proteomics, microRNA regulation, exposomics; network analysis
- **Programming and infrastructure**: R, Python (pandas, scikit-learn, PyTorch), Java, JavaScript, SQL, Bash; full-stack web application development; Nextflow and SLURM pipelines on Digital Research Alliance of Canada clusters; Git; containerized, reproducible workflows
- **Data resources**: UK Biobank, Million Veteran Program, Michigan Genomics Initiative, FinnGen, Canadian Longitudinal Study on Aging, CHILD Cohort Study, GTEx, UKB-PPP

Languages

Chinese (Mandarin; native) · English (full professional proficiency)